QP Code: 25901001

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION, NOVEMBER 2025
First Semester
Discipline Specific Core Course (DSC) –MG1DSCZGY100 , MG1DSCZIM100 –
INTRODUCTION TO ZOOLOGY
(2024 ADMISSION ONWARDS)

Time: 1.5 Hours **Max. Marks: 50**

Part A

Fill in the blanks
*Answer **All** questions*
*Each question carries **1 mark***

1. **In pearl culture, the small piece of tissue inserted inside the mantle is called as …(U)(CO1)**
   Graft tissue (Give mark if the answer is written as 'nucleus')

2. **The male humming birds provide indirect parental care by …….(U)(CO2)**
   guarding the nesting territory, aiding in nesting territory and feeding assistance. (Give mark for any of these points)

3. **The light in the anglerfish's lure is produced by …… living in the lure.(U)(CO1)**
   Bacteria

4. **…… is produced from dried and processed animal blood. (U)(CO3)**
   Blood meal

5. **…….. are the males of a bee colony.(U)(CO1)**
   Drones

6. **Increasing the body temperature by absorbing heat from the sun is called ……(U)(CO2)**
   Basking

7. **The behaviours and interactions of marine organisms with the environment is studied under ……(K)(CO3)**
   Marine Biology

8. **…….. is a common form of parental care in birds, where parents keep their eggs warm until they ……. (U)(CO2)**
   (a) Nesting (b) Hatch

9. **…… is the mechanism used by an organism to disguise their appearance and blend in with their surroundings.(U)(CO2)**
   Camouflage

10. **The energy for the bioluminescent reaction is provided by the ……molecule.(K)(CO1)**
    ATP

(10 x 1=10)

Part B

Short Questions

*Answer any **10** questions*

*Each question carries **2 marks***

11. Explain the by-products of Mulberry cultivation.(U)(CO4)

The discarded mulberry leaves may be dried in shades, and added to the silkworm's semi-synthetic diet. Due to the high nutritive value excess leaves may be used in cattle, rabbits, and chicken feeds. Mulberry tea, a concoction made from mulberry leaves is effective in lowering blood pressure. Dried up mulberry shoots is used as firewood and fuel. Due to high elasticity and flexibility mulberry wood is used for making furniture, sporting items and agricultural tools. The rejected mulberry plant material is used in paper industry. **(4 points-2 marks).**

12. Discuss how colouration helps in thermoregulation in animals.(U)(CO2)

Colouration influences the rate of absorption or reflection of solar radiation. Dark colour absorb more solar radiations and heat up faster, while light colour reflect more sunlight and heats up slowly. Melanin pigment also dark colour in animals. Animals can also change its body colour to regulate body temperature. Darkening the body colour helps to absorb more heat & lightening reflects heat.

13. What is a pearl? What is its composition?(U)(CO1)

Pearl is the calcareous concretion of a peculiar lusture secreted by a group of bivalve molluscs, the pearl oysters.

Composition:

- 85-90% Calcium Carbonate Crystals
- 5-7% Organic substances
- 4% water and other residues

14. Comment on Bee bread.(U)(CO4)

Bee bread is a fermented food product created by honeybees from pollen, nectar, and bee saliva, which is highly nutritious and protein rich fermented food source for the bees. It is rich in biologically active compounds, including vitamins, minerals, amino acids, proteins, lipids, fatty acids, enzymes, and antioxidants like flavonoids and carotenoids. It has anti-oxidant, anti-inflammatory, anti-tumour, anti-bacterial and anti-fungal properties.(**2 points-2 marks)**

15. List out the behavioural adaptations exhibited by water bug for parental care.(U)(CO2)

Male water bugs guard their eggs, groom them, actively fan & aerate and also clean them to maintain optimal temperature & conditions for development. In some species, males physically transport the eggs.

16. What is meant by neomelia?(U)(CO2)

It is a type of highly specialized parental care exhibited by Darwin frogs, where the males ingest the developing embryos & transport them in their vocal sacs until the metamorphosis is completed. During this time the males do not consume food. *Rhinoderma darwinii* also nourishes the tadpoles with viscous secretions within the vocal sac along with giving protection.

17. Explain why the velvet spider *Stegodyphus lineatus* reproduce only once in her entire lifetime?(U)(CO2)

The female velvet spider exhibits an extreme form of maternal care known as matriphagy, where she sacrifices her own body to feed her young. It is also known as suicidal maternal care, because the mother makes no attempt to escape her fate. Thus the mother's body becomes a vital

food source that gives the young necessary nutrients to survive their vulnerable early stages. Since they sacrifice their own body, they do not survive for the next reproductive act.

18. Define the phenomenon of bioluminescence.(K)(CO1)

Bioluminescence is light produced by a chemical reaction (chemiluminescence) within a living organism. Examples: firefly, *Noctiluca*, *Octopus*, Angler fish, some bacteria, fungi etc.

19. Describe the duties of a Wildlife Biologist/Wildlife Researcher.(U)(CO3)

Conduct field research on wildlife species, habitats and ecosystems. Collect and analyse data to contribute to conservation efforts and management plans. Collaborate with local communities and stakeholders to promote biodiversity conservation. **(Any Four)**

20. What is the significance of the mother otter's role in teaching her pups to swim and hunt? (U)(CO2)

Teaching survival skills is a crucial role played by mother otter. They gradually introduce their pups into water, initially carrying them on their back before independent swimming. They teach them swimming through playful interactions & supervised practice. Hunting skills are taught by bringing prey live prey to the den & demonstrating techniques, allowing the pups to practice & mimic these behaviours. Sea otters use tools to crack open shells. All these vital skills are taught by the mother, which in turn increases the survival chances of the young ones.

21. Dairy farming play a crucial role in rural economy. Substantiate the statement with a suitable example.(U)(CO4)

Dairy farming is the core activity within the diary sector, offering substantial employment opportunities especially in the rural area to numerous farmers and farm workers. Employment opportunities as Diary farm manager, Diary hard manager, stock men, milkers and milk testers are associated with diary farming. Opportunities are also linked in areas like, diary processing, marketing and distribution, cattle feed manufacturing units, Veterinary services, Field artificial Insemination services, Animal nutritionist and Agro-tourism area. Through all these areas, especially the people in the rural sectors will become engaged in the diary sectors that will definitely strengthen the economy.(**4 points- 2 marks)**

22. What materials do weaver birds use to build their nests?(U)(CO1)

Weaver birds use leaves of grass, reed, palm or bamboo to build their nests. They tear long strips from these leaves to form as threads. The birds prefer green grass for weaving as it is flexible and becomes strong when it dries up later. They also chooses a strong forked twig hanging downward from a tree as a support to weave a circle or hanging loop on which they build the nest.

(10 x 2 = 20)

Part C

Short Essay Type Questions
*Answer any **5** questions*
*Each question carries **4 marks***

23. Summarise the research avenues in pet behaviour and training and what career opportunities does this field offer?(U)(CO3)

Research avenues in pet behavior and training mainly focus on the influence of training on specific behaviors, effectiveness and impact of training methods and the welfare implications of different approaches.

Several entrepreneurship opportunities available with pet care management mainly in the areas of pet food making business, pet cloth accessories, pet sore, pet blog, pet grooming business, pet photography business, pet transportation services, pet boarding and day care and pet training and fitness. ***(Liberal Valuation).***

24. Describe how a new colony is formed in termites?(U)(CO1)

New colonies are established by winged reproductive adults, known as alates. Alates emerge from mature colonies and take flight during specific seasons to mate and create new nests. With the arrival of rainy season, these winged males and females swarm out of their nests at night. Many of them fall as prey to predators. A few alates will land, mate and initiate new nests. Upon landing the winged termites shed their wings. Each male follows a female until they find a suitable spot to dig a hole and start their nest. After mating, the newly formed king and Queen establish a life-long partnership, beginning the gradual process of building a new colony. They dig a small cell in the ground and the queen starts laying eggs. The King and Queen care for the eggs and young at first. When there are enough workers, they take over these jobs. The queen directs the workers to build the nest, using mud, soil, saliva and waste. The royal chamber houses the King and Queen, protected by soldiers. Nearby, the brood chambers serve as nurseries where workers care for eggs and nymphs, ensuring the colony's future workforce and defenders. (**Give full marks for any relevant points)**

25. Describe the different mechanisms employed by animals for faking.(U)(CO2)

a) By mimicking/ impersonate other animals – Eg: Mimic octopus mimic other animals like lionfish, flatfish etc.

b) Camouflage – blending with the surroundings for avoiding detection by predators. Eg: Leaf tailed Gecko

c) Brood parasitism – certain birds lay eggs in the nest of other birds. The host mistakes them as their biological offsprings and raises the offsprings. Eg Cuckoo.

d) Pretending dead – certain animals when threatened, they limp, emit foul odour, drool & play dead. Eg: Opposum.

e) Courtship display – male peacock spiders raise their colourful abdominal flap like a peacock's tail during courtship display.

f) Aggressive mimicry – Angler fish, uses their esca as lure to attract prey.

(Explanation of any 4 Points)

26. Recall the various fisheries research and marine biology institutes in India.(K)(CO3)

CMFRI- Central Marine Fisheries Research Institute, CIFT - Central Institute of Fisheries Technology, CIBA- Central Institute of Brackish water Aquaculture, CIFE -Central Institute of Fisheries Education, CIFRI- Central Inland Fisheries Research Institute, School of Marine Sciences-Cochin University of Science and Technology(CUSAT), Department of Aquatic Biology and Fisheries- University of Kerala, CMLRE-Centre for Marine Living Resources and Ecology, KUFOS- Kerala University of Fisheries and Ocean Sciences. (Any **4)**

27. Classify animal societies with example.(U)(CO1)

(a) Solitary Animals (e.g. Tiger.Leopards, Polar Bears)
(b) Pair-living Society (e.g. Albatross, Swans, Penguins, Fox)
(c) Family Group (e.g. Elephants, Lions, Wolves)
(d) Harems (e.g., Gorillas, Horses, Sea lions)
(e) Matriarchal Society (e.g., Elephants, Bonobos, Spotted heynas)
(f) Hierarchical Society (e.g., Chickens, Wolves, Baboons, Macaques)
(g) Egalitarian Societ (e.g., Bonobos, Some bird flocks)
(h) Fission-fusion Society (e.g., Chimpanzees, Dolphins, some bats)
(i) Co-operative Breeding Society (e.g., Meerkats, African Wild dogs, Acorn Woodpeckers)
(j) Eusocial Society (e.g., Honey bees, Ants, Termites)

(Any **8** points)

28. Explain how male seahorses protect and care for their eggs during the pregnancy period. (U)(C02)

Males carry the eggs in their brood pouch, which is deposited by the female with the help of her ovipositor. Males take care of the eggs throughout the incubation period, lasting from 10 days to 6 weeks. They maintain controlled environment within its pouch for embryonic development. It ensures optimal temperature & salinity inside the pouch, provide oxygen & nutrients directly through the walls of the pouch. The eggs are also protected from predators. Once the development is complete, they experience contractions & give birth to young ones. They invest much of their time and energy for nurturing their young ones.

29. Discuss the entrepreneurial opportunities in Vermiculture.(U)(CO4)

Entrepreneurship opportunities in vermiculture mainly focused on the areas like products and services, market expansion and business development and management. Youth empowerment through vermiculture enterprises include sustainable employment, supplemental income sources, rural –urban linkage, capacity building and training, market linkage (**Brief explanation)**

(5 X 4 = 20)

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